

Vessel Motion App

A Streamlit app for calculating vessel point vertical motion from RAO Excel workbooks.

Files

- `vessel_motion_app.py` — main Streamlit app
- `requirements.txt` — Python dependencies for Streamlit Community Cloud

Local run

```
python3 -m pip install --user -r requirements.txt
python3 -m streamlit run vessel_motion_app.py
```

Streamlit Community Cloud deploy

1. Create a GitHub repository.
2. Upload the files in this folder to the repository root.
3. Go to <https://share.streamlit.io>.
4. Connect your GitHub account.
5. Click **Create app**.
6. Select the repository and branch.
7. Set the main file path to `vessel_motion_app.py`.
8. Click **Deploy**.

Notes

- Upload the RAO workbook inside the app when prompted.
- The app parses multi-row grouped headers for Heave, Roll, and Pitch amplitude/phase columns.
- The app calculates RMS, significant, and most probable maximum vertical response metrics.
- The z coordinate is stored for reference, but the vertical rigid-body displacement uses heave, pitch with X offset, and roll with Y offset.